

Spin Aerodynamic Modeling for a Fixed-Wing Aircraft Using Flight Data

James L. Gresham,^{*} Benjamin M. Simmons,[†] Jeremy W. Hopwood,[‡] and Craig A. Woolsey[§]
Virginia Tech, Blacksburg, Virginia 24061

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Novel techniques are used to identify a nonlinear, quasi-steady, coupled, spin aerodynamic model for a fixed-wing aircraft from flight-test data. Orthogonal phase-optimized multisine inputs are used as excitation signals while collecting spinning flight data. A novel vector decomposition of explanatory variables leads to an elegant model structure for spin flight data analysis. Results show good agreement between model predictions and validation flight data. This effort is motivated by interest in developing a flight termination system for a fixed-wing unmanned aircraft that controls a descending spiral trajectory flight path toward a designated impact area. While investigating the feasibility of a robust control method to guide the spinning trajectory, it was helpful to compare a level flight dynamic model with one of the aircraft dynamics and control authority in the neighborhood of a stable, oscillatory spin. In this paper, a nominal flight aerodynamic model is developed and compared to the stall spin model and the spin model outperforms the nominal model for spinning flight.

Nomenclature

a_x, a_y, a_z	=	body-axis translational accelerations, ft/s ² or g
b	=	wingspan, ft
C_l, C_m, C_n	=	body-axis aerodynamic moment coefficients
C_x, C_y, C_z	=	body-axis aerodynamic force coefficients
$\bar{c}$	=	mean aerodynamic chord, ft
g	=	gravitational acceleration, ft/s ²
I_{xx}, I_{yy}, I_{zz}	=	moments of inertia, slug · ft ²
I_{xy}, I_{xz}, I_{yz}	=	products of inertia, slug · ft ²
$\mathcal{L}, \mathcal{M}, \mathcal{N}$	=	body-axis aerodynamic moments, ft-lbf
m	=	aircraft mass, slug
p, q, r	=	body-axis angular velocity components, rad/s or deg/s
$\bar{q}$	=	dynamic pressure, lbf/ft ²
S	=	wing area, ft ²
T	=	thrust, lbf
u, v, w	=	body-axis translational velocity components, ft/s
V_t	=	true airspeed, ft/s
X, Y, Z	=	body-axis aerodynamic forces, lbf
α	=	angle of attack, rad or deg
β	=	angle of sideslip, rad or deg
$\delta_a, \delta_e, \delta_r$	=	control surface deflections, rad or deg
$\hat{\theta}$	=	parameter estimate
ρ	=	air density, slug/ft ³
σ	=	parameter estimate uncertainty bound
ϕ, θ, ψ	=	Euler angles, rad or deg

I. Introduction

IDENTIFICATION of an aerodynamic model for stall and spin conditions is typically accomplished from data collected in a wind tunnel or using computational fluid dynamics (CFD). This study presents the methodology and results of identifying a nonlinear, quasi-steady, coupled aerodynamic model for a fixed-wing unmanned aircraft in a spinning descent from flight-test data [1]. The need for such a spin model arises in scenarios where one wishes to influence the helical trajectory through control, such as in Ref. [2]. For this paper, a flight-test experiment was designed and conducted using automated orthogonal phase-optimized multisine inputs [3,4] for excitation while in a stable spin. Additionally, nominal gliding flight data were collected with multisine excitations to develop a gliding flight aerodynamic model. Time-domain equation-error methods were used to develop the model structures and to estimate parameter values for both gliding and spinning flight. The developed aerodynamic models were then validated using flight-test data not used in the modeling process. The predictive performances for both the gliding flight model and spinning flight model were compared for spinning descent flight data.

Many modern unmanned aerial vehicles (UAVs), including the RQ-4 Global Hawk, are equipped with a flight termination sequence used in the event of an in-flight emergency, such as loss of propulsion or communication [5]. One common flight termination approach is to put the aircraft in an intentional spin, thereby giving a steady descent with a small impact radius and a relatively low impact speed as compared to a gliding or spiral descent. The main advantage of a spin descent is dissipation of energy in a stable, predictable, and controlled manner [6]. One major reason a vertical stall spin is an effective fail-safe is that it is typically a stable steady-state-periodic trajectory for fixed aileron, elevator, and rudder deflections. Some control authority is maintained in this state of motion because this authority is required to establish the motion, and thus it may be desirable to direct the spinning descent along a prescribed path to impact within a “safe” crash zone [7,8]. An aerodynamic model that describes aircraft motion in the neighborhood of a steady spin enables development and computational testing of a robust model-based control law that directs the spinning descent of an aircraft along a desired azimuth and inclination angle [2].

The primary contribution of this work is a methodology for developing a nonlinear, quasi-steady, coupled aerodynamic model for a fixed-wing unmanned aircraft in a spinning descent. This system identification methodology includes, to the best of the authors’ knowledge, the first use of multisine inputs as excitation signals while collecting spinning flight data and a novel decomposition of explanatory variables that leads to an elegant model structure to adequately describe spin dynamics using flight data. This paper uses techniques, theory, and processes developed in Refs. [2,9–12]. Reference [2]

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^{*}Postdoctoral Research Affiliate, Kevin T. Crofton Department of Aerospace and Ocean Engineering; also Flight Test Engineer and Lieutenant Colonel, U.S. Air Force, Arnold AFB, TN 37389. Senior Member AIAA.

[†]Ph.D. Candidate, Kevin T. Crofton Department of Aerospace and Ocean Engineering. Member AIAA.

[‡]Ph.D. Candidate, Kevin T. Crofton Department of Aerospace and Ocean Engineering. Student Member AIAA.

[§]Professor, Kevin T. Crofton Department of Aerospace and Ocean Engineering. Associate Fellow AIAA.

leveraged the identified spin model to develop parallel linear quadratic and robust H_∞ controllers that can adjust the direction of a spinning descent to improve predictability and safety during an intentional flight termination.

Prior research efforts on aircraft spin modeling were studied to gain insight into the current state of the associated theory, excitation techniques for model identification, analysis methods, and best practices. Fixed-wing aircraft spin behavior was studied intensively by the National Aeronautics and Space Administration (NASA) in the 1980s, with the primary aim of spin prevention and spin recovery in piloted aircraft. These studies were focused on spin characteristics and primarily based on rotary balance wind-tunnel data [13,14]. Subsequent NASA research blended vertical spin wind-tunnel data with forced oscillation wind-tunnel data to model aircraft spin dynamics [15]. Reference [16] studied the uncertainty of spin aerodynamic model parameter estimates using the blending techniques used in Ref. [15] for a large database of wind-tunnel experiments. Reference [17] demonstrated the importance of modeling unsteady effects for stall-spin departure dynamics. References [18,19] described the NASA 20-Foot Vertical Spin Tunnel and test methods for spin susceptibility, spin recovery, and stable spin characterization. Similar to the present paper, Refs. [20,21] each developed a nonlinear aerodynamic model for spinning flight using flight-test data alone. Reference [20] used local models partitioned based on the angle of attack and highlighted the importance of kinematic consistency of measured states for spin flight data. Reference [21] combined two novel modeling methods: the ensemble empirical mode decomposition technique, and the extended multi-point modeling approach. In contrast with the present paper, Refs. [20,21] did not excite the aircraft dynamics with control surface perturbations once in a stall spin. Reference [21] also incorporated numerous nonlinear model terms relative to the quantity of modeling flight data, and it did not present pairwise correlation of regressors.

This paper addresses a particular need for technical guidance concerning flight testing and data analysis for the identification of a flight dynamics model that adequately describes the response of a fixed-wing aircraft to control and other perturbations from a stable, oscillatory spinning motion. The remainder of the paper is organized as follows. Section II describes stable, steady spinning flight and the modeling process. Section III describes the aircraft flown in this effort and design of the flight-test experiment. Section IV presents results from flight tests and system identification. Section V discusses conclusions, as well as plans for ongoing work.

II. Background

A. Aerodynamics of a Spin

The spin condition for a fixed-wing aircraft in classical aviation is a rotating stall, which is “driven by adverse yaw and post-stall

lift differences” [22]. Aircraft with ailerons, as opposed to spoilers, experience adverse yaw from differential drag due to lift, which creates a yawing moment away from the direction of a roll input. If large aileron inputs are applied near the stall angle of attack without a sufficient rudder deflection to counteract the adverse yaw, an aircraft can enter a spin. This spin may progress to a stable spin without proper inputs to reduce the angle of attack below stall [22]. A spin may also be intentionally entered by approaching the stall angle of attack and then, at the onset of stall, applying a sustained large rudder input in the desired spin direction. Subsequent aileron inputs might reinforce the spin or counteract the steady spin. Of note, aircraft dynamics near stall, specifically considering applied yawing moments while simultaneously rolling, are complicated by the respective shift in the angle of attack on each wing (positive on the downward wing, and vice versa). These complex dynamics and possible schema for roll control were discussed in Ref. [23].

A classic spin is described by four phases: spin entry, incipient spin, steady (or developed) spin, and spin recovery [24]. The pilot techniques necessary for spin entry and spin recovery were required for this flight-test effort to enable modeling and predicting the aircraft dynamic response in the steady spin phase. A classic stable spin is characterized by steady motion with a constant angle of attack α , angle of sideslip β , spin rate Ω , and descent rate while the aircraft follows a vertical helical path. An oscillatory spin has a similar vertical helical path; however, α and β oscillate about their respective average values. The aircraft studied in this effort exhibits a stable oscillatory spin, as will be shown later in Sec. IV.

B. Dynamic Modeling of a Spin

The aerodynamic response of a fixed-wing aircraft to small perturbations from steady, level flight at a low angle of attack is well characterized by a linear aerodynamic force and moment model, $f(\alpha, \beta, p, q, r, \delta_e, \delta_a, \delta_r)$, where the equations of motion decouple into longitudinal and lateral-directional components. Aggressive out-of-plane flight maneuvers such as a stall spin are best described by a nonlinear flight dynamic model where the equations of motion are fully coupled due to nonlinear aerodynamics and rotating reference frame effects. A nonlinear flight dynamic model is also needed when describing dynamic perturbations from a given steady motion outside of the prestall envelope, such as a stable spin.

For this effort, with inspiration from Refs. [13–15], the flight data recorded onboard the aircraft are transformed into states better suited to characterizing spinning motion. The body-axis angular velocity ω_b is decomposed, as depicted in Fig. 1, into components along (ω_{nom}) and orthogonal to (ω_{osc}) the instantaneous relative wind vector:

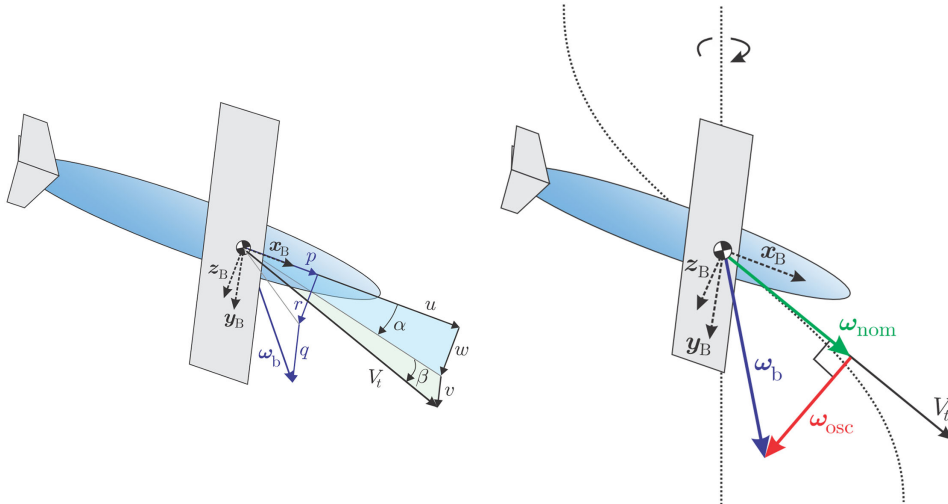


Fig. 1 Spin dynamic motion notation and decomposition of body-axis rotation rates in a perturbed spin.

$$\omega_b = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{\text{body}}, \quad \omega_{\text{nom}} = \Omega \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}_{\text{wind}} = \Omega \begin{bmatrix} \cos \alpha \cos \beta \\ \sin \beta \\ \sin \alpha \cos \beta \end{bmatrix}_{\text{body}}, \quad (1)$$

$$\omega_{\text{osc}} = \begin{bmatrix} p_{\text{osc}} \\ q_{\text{osc}} \\ r_{\text{osc}} \end{bmatrix}_{\text{body}}$$

In this notation, ω_{nom} is the projection of ω_b onto the instantaneous relative wind vector. Because the body-axis angular velocity vector and relative wind vector share an origin at the center of gravity, ω_{nom} has only a single component in the wind frame, which is the angular rate about the relative wind vector, denoted Ω . The orthogonal complement ω_{osc} comprises three rates, which are denoted as the body-axis oscillatory components because they describe residual body-axis oscillations that are not directly about the instantaneous relative wind vector. For a steady-state spin, Ω in Eq. (1) is the classical spin rate and $\omega_{\text{osc}} = \mathbf{0}$. This notation is relatively consistent with that used for blending vertical spin tunnel rates and forced oscillation rates from wind-tunnel data [15,25]. The rate components are calculated from

$$\Omega = p \cos \alpha \cos \beta + q \sin \beta + r \sin \alpha \cos \beta \quad (2)$$

$$p_{\text{osc}} = p - \Omega \cos \alpha \cos \beta \quad (3)$$

$$q_{\text{osc}} = q - \Omega \sin \beta \quad (4)$$

$$r_{\text{osc}} = r - \Omega \sin \alpha \cos \beta \quad (5)$$

The measured body-axis angular rates projected onto the relative wind vector and the residual oscillatory components are used during the model structure determination phase of system identification. This transformation, based on aerodynamic spin theory, results in significantly improved modeling results.

C. Aircraft System Identification

System identification is the process of developing a mathematical model of a physical system based on observed data from the system [26]. The objective of system identification is to create a *useful* system model based on measured or calculated *inputs* and *outputs*. The system identification process is composed of multiple steps that typically result in a set of differential or algebraic equations describing system behavior. Aircraft system identification seeks to develop an aerodynamic model that adequately describes the aircraft motion or response (outputs) to controls or perturbations (inputs). System identification is performed using imperfect knowledge of the inputs and outputs due to sensor noise, process noise, sensor scale factors, and bias that must be accounted for during experiment design and analysis. The aircraft system identification methodology, inspired by Ref. [3] and adapted for this small fixed-wing UAV spin research, is shown in Fig. 2.

For this work, data compatibility (or kinematic consistency) analysis was accomplished using both integration and smoothed differentiation of kinematic state estimates and measurements. Next, the model structure was determined from experimental data using multivariate orthogonal function (MOF) modeling [3,27]. Values for the unknown model parameters were determined using the equation-error method [3]. Finally, the model was validated using flight data that were not used in the modeling process. In general, the “independent data” indicated in Fig. 2 might include CFD or wind-tunnel data. For this effort, the independent data were collected in flight using a multisine excitation that was developed with a different power spectral density and a different time trace for each control surface excitation signal than the multisine excitation used for collecting data used for model development.

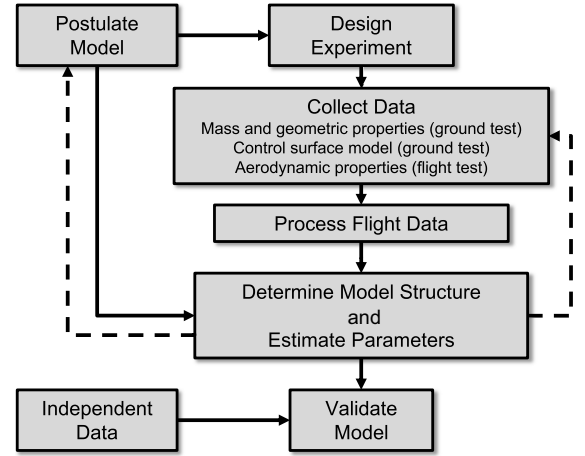


Fig. 2 Aircraft system identification process for spin aerodynamic modeling (adapted from Ref. [3]).

System identification from flight data requires low pairwise correlation between individual explanatory variables in order to identify the contribution of each variable to the aircraft dynamic response. One excitation technique used for aircraft system identification is the orthogonal phase-optimized multisine input [3,4]. This technique offers the primary advantage of providing rich excitation while minimizing test time. The technique ensures low correlation between the control inputs, which improves the modeling results. The design of multisine inputs does not require *a priori* information of the aircraft dynamics, but it does require precisely scripted, automated control inputs [28]. For this effort, orthogonal multisine inputs were developed and applied to the elevator, aileron, and rudder control surfaces to excite aircraft motion about all three axes concurrently. Design of the multisine inputs for this paper will be discussed further in Sec. III.B. The efficient excitation offered by multisine inputs is well suited to the short-duration stall spin maneuvers performed in this research. Additionally, well designed and implemented multisine signals are optimized to keep the aircraft near the starting flight condition, which for this study allowed the aircraft to remain in, and not break out of, the steady spin condition. Finally, with the excitation inputs automated, the pilot could focus on maintaining spin inputs and simply activate the automated excitations. Excitation of dynamics in spinning flight with remote uncorrelated pilot inputs, as described in Ref. [10], was considered for this effort; however, the modeling results were improved by using automated multisine excitations.

D. Aerodynamic Model Structure

A model framework is initially postulated from physical principles to aid in designing the experiment, and then the final aerodynamic model structure is determined based on aerodynamic phenomena reflected in experimental data. As seen in Fig. 2, the model structure development may be an iterative process, which further guides the experiment design. The model framework takes the form of the rigid-body equations of motion (EOMs). The aircraft was modeled as a rigid body of mass m with known principal moments of inertia I_{xx} , I_{yy} , and I_{zz} , as well as product of inertia I_{xz} . The remaining products of inertia, I_{xy} and I_{yz} , were assumed to be zero based on lateral symmetry.

The six-degree-of-freedom body-axis EOMs are as follows:
Rotational kinematic equations:

$$p = -\dot{\psi} \sin \theta + \dot{\phi} \quad (6)$$

$$q = \dot{\psi} \sin \phi \cos \theta + \dot{\theta} \cos \phi \quad (7)$$

$$r = \dot{\psi} \cos \phi \cos \theta - \dot{\theta} \sin \phi \quad (8)$$

Translational dynamics (force) equations:

$$X = m(\dot{u} - rv + qw + g \sin \theta) \quad (9)$$

$$Y = m(\dot{v} + ru - pw - g \sin \phi \cos \theta) \quad (10)$$

$$Z = m(\dot{w} - qu + pv - g \cos \phi \cos \theta) \quad (11)$$

Rotational dynamics (moment) equations:

$$\mathcal{L} = \dot{p}I_{xx} + qr(I_{zz} - I_{yy}) - (\dot{r} + pq)I_{xz} \quad (12)$$

$$\mathcal{M} = \dot{q}I_{yy} - pr(I_{zz} - I_{xx}) + (p^2 - r^2)I_{xz} \quad (13)$$

$$\mathcal{N} = \dot{r}I_{zz} + pq(I_{yy} - I_{xx}) + (qr - \dot{p})I_{xz} \quad (14)$$

The aerodynamic forces (X , Y , and Z) and moments ($\mathcal{L}$, $\mathcal{M}$, and $\mathcal{N}$) are expressed in the nondimensional forms:

$$C_x = \frac{X}{\bar{q}S}, \quad C_y = \frac{Y}{\bar{q}S}, \quad C_z = \frac{Z}{\bar{q}S} \quad (15)$$

$$C_l = \frac{\mathcal{L}}{\bar{q}Sb}, \quad C_m = \frac{\mathcal{M}}{\bar{q}S\bar{c}}, \quad C_n = \frac{\mathcal{N}}{\bar{q}Sb} \quad (16)$$

to remove dependence on the dynamic pressure ($\bar{q} = \rho V_i^2/2$) and aircraft size.

Aircraft system identification is accomplished by obtaining expressions for the aerodynamic force and moment coefficients [3,29]. Aerodynamic force coefficients are estimated using the mass m , dynamic pressure $\bar{q}$, body-frame measurements of translational acceleration (a_x , a_y , and a_z), and thrust T , which is assumed to act along the body x axis:

$$C_x = \frac{ma_x - T}{\bar{q}S} \quad (17)$$

$$C_y = \frac{ma_y}{\bar{q}S} \quad (18)$$

$$C_z = \frac{ma_z}{\bar{q}S} \quad (19)$$

This research involved maneuvers with idle propulsion; thus, the separate thrust model is omitted, and windmilling propeller effects are included in the aerodynamic model. The aerodynamic moment coefficients are calculated by combining the nondimensional moment equations shown in Eq. (16) and the rotational dynamics equations [Eqs. (12–14)], leading to

$$C_l = \frac{1}{\bar{q}bS} \left(I_{xx}\dot{p} - I_{xz}(\dot{r} + pq) + qr(I_{zz} - I_{yy}) \right) \quad (20)$$

$$C_m = \frac{1}{\bar{q}\bar{c}S} \left(I_{yy}\dot{q} + rp(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \right) \quad (21)$$

$$C_n = \frac{1}{\bar{q}bS} \left(I_{zz}\dot{r} - I_{xz}(\dot{p} - qr) + pq(I_{yy} - I_{xx}) \right) \quad (22)$$

Direct measurements of angular acceleration are unavailable, and so these terms are obtained as smoothed numerical derivatives of the angular rate measurements [3].

III. Flight-Test Experiment

A. Research Aircraft and Instrumentation

The radio-controlled (RC) fixed-wing aircraft selected for testing was the E-flite Carbon-Z Cessna 150 (CZ150), shown in Fig. 3. The



Fig. 3 CZ150 fixed-wing research aircraft.

Table 1 The CZ150 mass and geometric properties

Property	Symbol	Value	Units
Mass	m	0.336	slug
Mean aerodynamic chord	$\bar{c}$	1.049	ft
Wingspan	b	6.97	ft
Wing area	S	7.31	ft ²
Aspect ratio	AR	6.65	—
Roll moment of inertia	I_{xx}	0.457	slug · ft ²
Pitch moment of inertia	I_{yy}	0.357	slug · ft ²
Yaw moment of inertia	I_{zz}	0.622	slug · ft ²
Product of inertia	I_{xz}	0.052	slug · ft ²

aircraft is electrically powered by a six-cell 22.2 V, 5000 mAh lithium-polymer battery, with a tractor propeller configuration using a single 16-in. diameter, 8-in. pitch (16 × 8) Aeronaut CAM Carbon folding propeller. The CZ150 aircraft was instrumented with a Pixhawk CubePilot Cube Orange flight controller for data collection running PX4 firmware.^{†, **} Additionally, an onboard companion computer was used for automated input generation. The companion computer was a Raspberry Pi 4 running Ubuntu Linux distribution communicating with the Pixhawk using Mavlink protocols and using the Robot Operating System libraries. The Pixhawk sensor suite includes an attitude and heading reference system and a real-time kinematic Global Positioning System receiver. Details concerning postflight data processing and companion computer implementation are presented in Refs. [11,12].

The CZ150 was chosen primarily for its ability to enter and maintain a spin, as well as its benign spin recovery characteristics. Additional advantages were its simple construction, satisfactory handling qualities, and sufficient payload capacity to incorporate the desired instrumentation package and onboard companion computer. The CZ150 mass and geometric properties, tabulated in Table 1, were determined with methods described in Ref. [30]. The principal moments of inertia were measured experimentally using the compound pendulum technique [31], and the product of inertia I_{xz} was determined using a test described in Ref. [3]. The remaining products of inertia, I_{xy} and I_{yz} , were assumed to be zero based on lateral symmetry.

The Pixhawk PX4 firmware includes an extended Kalman filter (EKF) state estimator that provides orientation angles and Earth-fixed velocity (see footnote **). The aircraft instrumentation system lacked air data angle sensors; therefore, α and β were derived from the EKF states. The kinematic consistency analysis to be presented in Sec. IV shows the derived values of α and β are suitable for use in a stall spin. The pulse-width modulation signal sent from the Pixhawk to each control surface servo was used to determine the control surface deflection angle based on a servo-actuator model. Further discussion of the servo-actuator modeling techniques is presented in Ref. [12].

[†]“Cube Orange Flight Controller,” https://docs.px4.io/master/en/flight_controller/cubepilot_cube_orange.html [retrieved May 2021].

^{**}“Using the Estimation and Control Library Extended Kalman Filter,” https://docs.px4.io/master/en/advanced_config/tuning_the_ecl_ekf.html [retrieved February 2021].

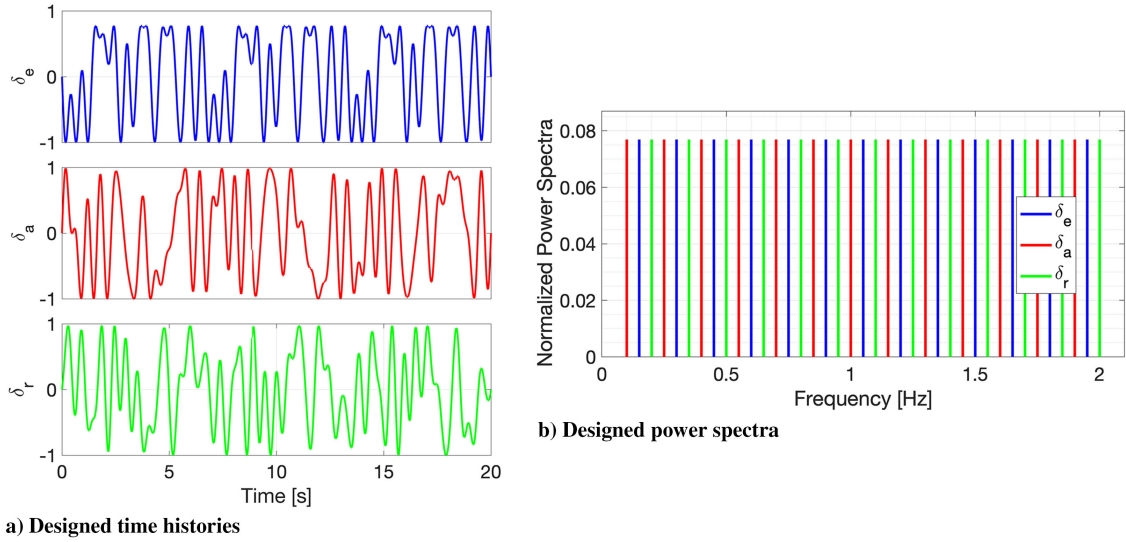


Fig. 4 Designed multisine control inputs.

B. Multisine Excitation Design

As discussed in Sec. II.C, orthogonal phase-optimized multisine inputs were designed to develop a comprehensive aircraft dynamic model around a nominal flight condition that described nonlinear aerodynamic phenomena and control surface interaction effects. A multisine input is defined as a sum of multiple sinusoidal functions with different amplitudes, frequencies, and phase angles, where the frequencies are chosen to excite the system dynamics of interest. To make all inputs orthogonal in both the time domain and frequency domain, the multisine signal for the j th control effector is assigned sinusoids with a unique subset K_j of discrete harmonic frequency indices selected from the complete set of K available frequency indices. The available frequencies are $f_k = k/T$, $k = 1, 2, \dots, K$, where T is the maneuver duration. For m total control effectors, the j th input signal u_j is defined as

$$u_j = \sum_{k \in K_j} A_j \sqrt{P_k} \sin\left(\frac{2\pi kt}{T} + \phi_k\right) \quad j = 1, 2, \dots, m \quad (23)$$

where A_j is the signal amplitude; P_k is the k th power fraction, with $\sum_{k \in K_j} P_k = 1$; ϕ_k is the k th phase angle defined on the interval $(-\pi, +\pi]$; and t is the time vector containing N discrete points. The phase angles are optimized to obtain a minimum relative peak factor, thereby helping to keep the system close to its nominal flight trajectory while also retaining the same high-input energy as a signal without optimized phase angles. After phase angle optimization, phase shifts are added so that the periodic multisine signal starts and ends at a value of zero [32].

Three-axis multisine inputs were designed for a range of 0.1 to 2.0 Hz using the System IDentification Program for Aircraft (SID-PAC) [3], which covered the frequency range of the dynamic modes of interest for gliding flight and appropriately excited the dynamics in the region of interest for spinning flight. The multisine inputs were designed for the three aerodynamic control inputs (elevator, aileron, and rudder) with a uniform power spectrum and 39 total discrete frequencies and phase angles. The designed time histories and power spectra are shown in Fig. 4, and the designed harmonics are listed in Table 2. The amplitude of each multisine input was normalized to unity and then adjusted in real time by the pilot with a knob from 0 to 100% of the available control surface deflection angle. Further details on implementation of multisine inputs during this flight-test experiment are discussed in the following sections.

C. Flight-Test Experiment Design

Flight-test data collection was accomplished over a series of flights at the Kentland Experimental Aerial Systems Laboratory located at

Table 2 Model development orthogonal phase-optimized multisine harmonics ($T = 20$ s)

f_{δ_e} , Hz	ϕ_{δ_e} , rad	f_{δ_a} , Hz	ϕ_{δ_a} , rad	f_{δ_r} , Hz	ϕ_{δ_r} , rad
0.15	+1.35	0.10	-0.47	0.20	-3.09
0.30	+1.20	0.25	+1.29	0.35	-1.55
0.45	-0.59	0.40	-2.30	0.50	+3.10
0.60	-2.02	0.55	-2.25	0.65	-1.72
0.75	+0.27	0.70	-0.49	0.80	+0.43
0.90	+1.96	0.85	+3.02	0.95	+1.38
1.05	-1.93	1.00	+0.83	1.10	-2.25
1.20	-0.75	1.15	+2.95	1.25	-2.77
1.35	-2.06	1.30	+2.31	1.40	+1.94
1.50	+0.33	1.45	+1.63	1.55	+1.53
1.65	+0.14	1.60	-1.74	1.70	-2.61
1.80	+1.21	1.75	-2.81	1.85	+1.47
1.95	-0.57	1.90	+3.13	2.00	-0.75

Virginia Tech's Kentland Farm agricultural research facility^{††}. The nominal glide model was developed from a collection of eight nominal gliding maneuvers with multisine excitations active on each control surface. Each maneuver included 1 s of trimmed flight followed by 30 s of excitation. The spin model was developed from a collection of 18 spin maneuvers with multisine excitations active on each control surface. For each spin maneuver, the pilot progressed through a series of scripted test inputs:

- 1) Initiate benign, wings-level, power-off stall (δ_e : 70% trailing-edge up gradually, δ_a and δ_r : as required to level wings).
- 2) At stall break, initiate spin inputs (δ_r : 70% left, δ_e : maintain 70% trailing-edge up, δ_a : 50% left).
- 3) After two spin rotations, initiate multisine excitations.
- 4) At predetermined safe recovery altitude, terminate the multisine excitations and recover to resume normal flight.

Precise and repeatable control surface deflections were obtained during test maneuvers by modifying the RC transmitter control logic. Before initiating spin maneuvers in flight, the pilot adjusted a switch that reduced the maximum transmitted control signal for each control surface to those values shown in step 2 of the aforementioned scripted test inputs (δ_e : 70%, δ_a : 50%, and δ_r : 70% of maximum values). This allowed the pilot to maintain a precise spin by holding the maximum control inputs (against the stops) on the RC transmitter. Actual control deflections onboard the aircraft, which were pilot inputs summed with automated excitations, did not saturate the control surface actuators. Each spin maneuver used for system identification

^{††}“Kentland Experimental Aerial Systems (KEAS) Laboratory,” <https://www.aoe.vt.edu/research/facilities/keas.html> [retrieved April 2021].

included only the data involving multisine excitations, after development of a steady spin. The useful data record was typically 7 to 12 s in length, depending on the initial and final altitudes. The short maneuver length and irregularities in the data due to the nonlinear flight regime made more maneuvers desirable for spin modeling than for gliding flight modeling.

D. Oscillatory Spin

During each flight executed to collect spin modeling data, a nominal spin without excitation inputs was accomplished as the first maneuver. This was primarily for safety consideration to demonstrate repeatability of the steady spinning motion and the four spin phases, including benign spin recovery characteristics. This unexcited spin maneuver also permitted collection of flight data to confirm the nominal spin conditions were comparable to spin maneuvers with multisine excitations. The flight path from a nominal spin flight maneuver is shown in Fig. 5a, illustrating the entry and initial development of the incipient spin, the fully developed steady spinning motion, and a successful recovery. The aircraft orientation is shown at a 1 s intervals with the flight path displayed based on the onboard data recorded at 50 Hz. Of note, the idle-power wings-level stall angle of attack for the CZ150 test aircraft was around 14 deg, as determined from the repeatable spin entry flight data. An α - β trace from the maneuver is shown in Fig. 5b with the oscillatory spin clearly evident, as discussed in Sec. II.A. Nominal spin data were collected with a 70% left rudder deflection, 70% trailing-edge up elevator, and 50% aileron deflection for left roll input. These inputs were selected to ensure the controls were not saturated with the multisine inputs active and that the aircraft remained in a stall spin during multisine excitations. Additionally, the selected nominal control inputs permit actuator deflections from Ref. [2] to control the descent path while maintaining a spin. The aerodynamic spin model developed is valid around a perturbation range from this nominal spin. Steady nominal (median) spin state and control input values determined from flight data, along with the fifth to 95th percentile limits for the collected modeling data, are shown in Table 3. Centered explanatory variables, using the nominal (median) values, were used for spin model structure determination, parameter estimation, and model validation.

IV. Results

The aircraft system identification methodology presented in Sec. II.C was followed to produce a nonlinear, coupled aerodynamic model for both nominal glide and spinning flight. The models were developed using only flight-test data without excessive disturbances, data dropouts, or poor kinematic consistency. Modeling was performed using flight data with multisine excitations in a nominal glide and a stable spin, as well as using the techniques and processing tools discussed in Sec. II. Validation was accomplished using data from

Table 3 Spin data ranges of validity and nominal (median) values

Factor	Units	Fifth percentile	Median	95th percentile
V_t	ft/s	56.9	62.1	70.5
ϕ	deg	-50.5	-25.9	-12.6
θ	deg	-83.0	-73.8	-64.5
α	deg	3.0	13.8	24.9
β	deg	-4.8	6.8	18.0
$\dot{\alpha}$	deg/s	-84.4	1.4	77.9
$\dot{\beta}$	deg/s	-79.6	-0.2	77.0
p	deg/s	-479.4	-411.0	-304.7
q	deg/s	-21.2	22.5	64.4
r	deg/s	-140.1	-111.8	-61.1
Ω	deg/s	-490.6	-416.3	-296.9
p_{osc}	deg/s	-52.0	-14.7	7.0
q_{osc}	deg/s	-17.5	71.4	143.5
r_{osc}	deg/s	-88.2	-11.9	52.3
δ_e	deg	-23.9	-16.9	-11.0
δ_a	deg	9.4	15.3	21.3
δ_r	deg	14.6	25.1	28.2

independently designed multisine input maneuvers. Data analysis was conducted in MATLAB® [33] with the use of analytic tools developed by the authors and those provided in SIDPAC [3]. To remain nondimensional, all angle calculations were accomplished in radians, and the angle-based derivative values shown in this section are expressed per radian.

A. Data Compatibility Analysis and Corrections

Figure 6 shows a data compatibility analysis plot for a spin maneuver with multisine excitations. The figure shows the corrected data used for system identification analysis, raw data obtained from the instrumentation system, and flight-path reconstruction (FPR) signals calculated from other measurements to confirm validity of the kinematic consistency corrections [3]. The reconstructed measurements and data processing, including smoothing and correction for sensor bias and time skews, were accomplished as discussed in detail in Ref. [12]. The state variable time lags were estimated visually via reconstruction, and the control lags were adjusted in the servo-actuator modeling process. After correction, the corrected and reconstructed signals showed excellent agreement, indicating that the data have been well conditioned to proceed with system identification. Notably, the derived α and β data obtained from the oscillatory spin agreed very well with the reconstructed signals.

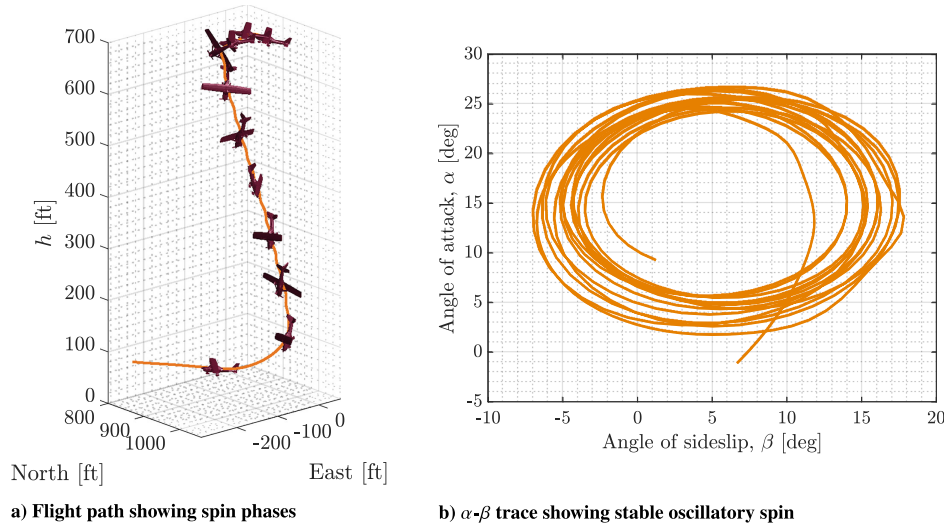


Fig. 5 Experimental flight data from a nominal spin maneuver.

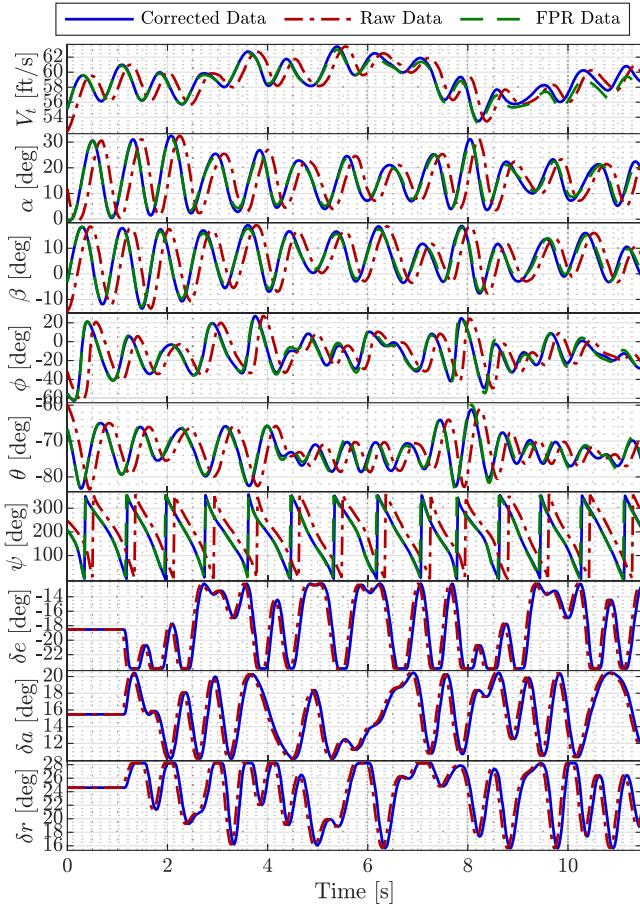


Fig. 6 Data compatibility for a spin maneuver with multisine excitations.

B. Model Structure

For this research, MOF modeling [3,27] was employed for initial model structure determination using the same steps described in Ref. [10], and then analyst insight was used to refine the model structure as described in this section. The cutoff for modeling terms was determined automatically using both the predicted squared error and coefficient of determination R^2 values, following the guidelines presented in Ref. [34].

The explanatory variables used to represent the aerodynamic force and moment coefficients for the nominal glide in this research were α , β , the nondimensional body-axis angular rate components

$$\hat{p} = \frac{pb}{2V_t}, \quad \hat{q} = \frac{qc}{2V_t}, \quad \hat{r} = \frac{rb}{2V_t}$$

and the control surface angular deflections δ_e , δ_a , and δ_r . These explanatory variables facilitated convenient control law tuning, and the nominal glide model developed with these variables produced a good match with validation data.

Initially, the same explanatory variables used to develop the nominal glide model were considered for spin modeling; however, this provided an inadequate model structure to describe the spin dynamics. As described in Sec. II, the body-axis recorded rates were transformed to the wind-axis angular rate Ω and the complementary body-axis oscillatory components (p_{osc} , q_{osc} , and r_{osc}) using Eqs. (2–5). Furthermore, based on demonstrated spin model improvement by including unsteady effects in Ref. [17], terms $\dot{\alpha}$ and $\dot{\beta}$ were considered, which significantly improved the predictive performance of the model. Thus, the explanatory variables used to represent the aerodynamic force and moment coefficients for spinning motion were α , β , their nondimensional rates

$$\hat{\alpha} = \frac{\dot{\alpha} \bar{c}}{2V_t}, \quad \hat{\beta} = \frac{\dot{\beta} b}{2V_t}$$

the nondimensional spin angular rates

$$\hat{\Omega} = \frac{\Omega b}{2V_t}, \quad \hat{p}_{osc} = \frac{p_{osc} b}{2V_t}, \quad \hat{q}_{osc} = \frac{q_{osc} \bar{c}}{2V_t}, \quad \hat{r}_{osc} = \frac{r_{osc} b}{2V_t}$$

and the control surface angular deflections δ_e , δ_a , and δ_r .

Figure 7 is a plot matrix showing correlation between the explanatory variables, following the example of Ref. [35], for a single spin maneuver with multisine excitations. The explanatory variables are labeled along the diagonal, pairwise correlations are shown in the lower triangle, and corresponding pairwise cross plots are shown diagonally opposite in the upper triangle. Low correlation between most explanatory variables is observed; however, the pairs $\dot{\alpha}/q_{osc}$, $\dot{\beta}/r_{osc}$, $\dot{\alpha}/\beta$, α/β , and α/r_{osc} are highly correlated. The MOF modeling process prioritizes including regressors that cause the largest decrease in the mean square fit error (MSFE) [3]. MOF modeling results indicated that some highly correlated regressor pairs should be included in the model structure because their independent contributions to modeling the response variable (i.e., associated MSFE decrease) warrant that both terms be included. Furthermore, flight data analysis showed model predictive performance notably improved when both terms were included.

MOF modeling was used with analyst insight to study various model structures. Candidate regressors were composed of the explanatory variables in all polynomial combinations up to the third order. Transcendental functions and spline models of α and β were considered, but they did not improve the results, likely because the aircraft remained within the poststall aerodynamic regime throughout the spin. Linear terms and lower-order terms were prioritized in accordance with model hierarchy. Finally, following parameter estimation with the equation-error method, terms were removed that had an aggregate median and mean near zero, with nearly equal runs having positive and negative parameter estimates.

For the nominal glide, the identified model structure for the force and moment coefficients is given in Eqs. (24–29). The model structure is valid in the normal operating range, including the onset of stall, but it does not capture poststall effects or spin dynamics. For ease of comparison, regressors common to both the nominal glide and spin models are colored red:

$$C_x = C_{x_\alpha} \alpha + C_{x_q} \hat{q} + C_{x_{\delta_e}} \delta_e + C_{x_{(\alpha^2)}} (\alpha^2) + C_{x_{(\beta^2)}} (\beta^2) + C_{x_o} \quad (24)$$

$$C_y = C_{y_\beta} \beta + C_{y_p} \hat{p} + C_{y_r} \hat{r} + C_{y_{\delta_a}} \delta_a + C_{y_{\delta_r}} \delta_r + C_{y_o} \quad (25)$$

$$C_z = C_{z_\alpha} \alpha + C_{z_q} \hat{q} + C_{z_{\delta_e}} \delta_e + C_{z_{(\beta^2)}} (\beta^2) + C_{z_o} \quad (26)$$

$$C_l = C_{l_\beta} \beta + C_{l_p} \hat{p} + C_{l_r} \hat{r} + C_{l_{\delta_a}} \delta_a + C_{l_o} \quad (27)$$

$$C_m = C_{m_\alpha} \alpha + C_{m_q} \hat{q} + C_{m_{\delta_e}} \delta_e + C_{m_{(\alpha^2)}} (\alpha^2) + C_{m_{(\alpha^3)}} (\alpha^3) + C_{m_{(\beta^3)}} (\beta^3) + C_{m_o} \quad (28)$$

$$C_n = C_{n_\beta} \beta + C_{n_p} \hat{p} + C_{n_r} \hat{r} + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r + C_{n_{(\beta^2)}} (\beta^2) + C_{n_{(\beta^3)}} (\beta^3) + C_{n_o} \quad (29)$$

For the steady spins, the identified model structure for each force and moment coefficient is given in Eqs. (30–35):

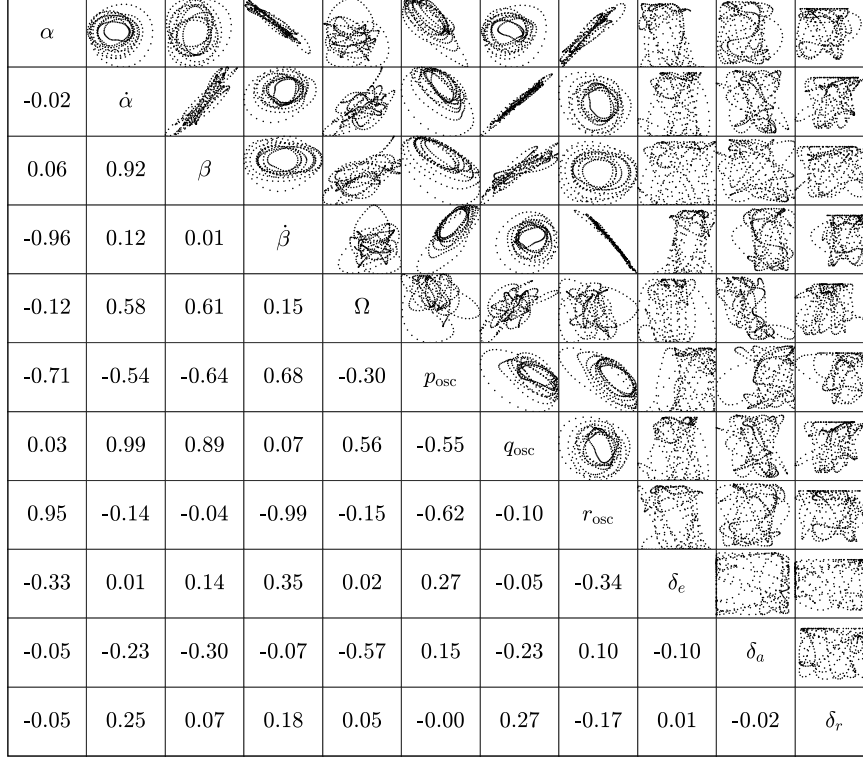


Fig. 7 Explanatory variable cross-correlation coefficients and cross plots for spin maneuver.

$$C_x = C_{x_\alpha} \alpha + C_{x_{\dot{\alpha}}} \dot{\alpha} + C_{x_\beta} \beta + C_{x_{\dot{\beta}}} \dot{\beta} + C_{x_\Omega} \Omega + C_{x_{p_{osc}}} \hat{p}_{osc} + C_{x_{q_{osc}}} \hat{q}_{osc} + C_{x_{r_{osc}}} \hat{r}_{osc} + C_{x_{\delta_e}} \delta_e + C_{x_{\delta_a}} \delta_a + C_{x_{\delta_r}} \delta_r + C_{x_{(\delta_e^2)}} (\delta_e^2) + C_{x_o} \quad (30)$$

$$C_y = C_{y_\alpha} \alpha + C_{y_{\dot{\alpha}}} \dot{\alpha} + C_{y_\beta} \beta + C_{y_{\dot{\beta}}} \dot{\beta} + C_{y_\Omega} \Omega + C_{y_{p_{osc}}} \hat{p}_{osc} + C_{y_{q_{osc}}} \hat{q}_{osc} + C_{y_{r_{osc}}} \hat{r}_{osc} + C_{y_{\delta_e}} \delta_e + C_{y_{\delta_a}} \delta_a + C_{y_{\delta_r}} \delta_r + C_{y_{(\delta_e^2)}} (\delta_e^2) + C_{y_o} \quad (31)$$

$$C_z = C_{z_\alpha} \alpha + C_{z_{\dot{\alpha}}} \dot{\alpha} + C_{z_\beta} \beta + C_{z_{\dot{\beta}}} \dot{\beta} + C_{z_\Omega} \Omega + C_{z_{p_{osc}}} \hat{p}_{osc} + C_{z_{q_{osc}}} \hat{q}_{osc} + C_{z_{r_{osc}}} \hat{r}_{osc} + C_{z_{\delta_e}} \delta_e + C_{z_{\delta_a}} \delta_a + C_{z_{\delta_r}} \delta_r + C_{z_{(\delta_e^2)}} (\delta_e^2) + C_{z_{(\delta_e \delta_a)}} (\delta_e \delta_a) + C_{z_o} \quad (32)$$

$$C_l = C_{l_\alpha} \alpha + C_{l_{\dot{\alpha}}} \dot{\alpha} + C_{l_\beta} \beta + C_{l_{\dot{\beta}}} \dot{\beta} + C_{l_\Omega} \Omega + C_{l_{p_{osc}}} \hat{p}_{osc} + C_{l_{q_{osc}}} \hat{q}_{osc} + C_{l_{r_{osc}}} \hat{r}_{osc} + C_{l_{\delta_e}} \delta_e + C_{l_{\delta_a}} \delta_a + C_{l_{\delta_r}} \delta_r + C_{l_{(\delta_e^2)}} (\delta_e^2) + C_{l_o} \quad (33)$$

$$C_m = C_{m_\alpha} \alpha + C_{m_{\dot{\alpha}}} \dot{\alpha} + C_{m_\beta} \beta + C_{m_{\dot{\beta}}} \dot{\beta} + C_{m_\Omega} \Omega + C_{m_{p_{osc}}} \hat{p}_{osc} + C_{m_{q_{osc}}} \hat{q}_{osc} + C_{m_{r_{osc}}} \hat{r}_{osc} + C_{m_{\delta_e}} \delta_e + C_{m_{\delta_a}} \delta_a + C_{m_{\delta_r}} \delta_r + C_{m_{(\delta_e^2)}} (\delta_e^2) + C_{m_{(\delta_e \delta_a)}} (\delta_e \delta_a) + C_{m_o} \quad (34)$$

$$C_n = C_{n_\alpha} \alpha + C_{n_{\dot{\alpha}}} \dot{\alpha} + C_{n_\beta} \beta + C_{n_{\dot{\beta}}} \dot{\beta} + C_{n_\Omega} \Omega + C_{n_{p_{osc}}} \hat{p}_{osc} + C_{n_{q_{osc}}} \hat{q}_{osc} + C_{n_{r_{osc}}} \hat{r}_{osc} + C_{n_{\delta_e}} \delta_e + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r + C_{n_{(\delta_e^2)}} (\delta_e^2) + C_{n_o} \quad (35)$$

The model captures longitudinal and lateral-directional coupling, poststall effects, and spin dynamics. The model structure

is valid around a perturbation range centered at the nominal spin for 70% left rudder deflection, 70% trailing-edge up elevator, and 50% aileron deflection for left roll input, as described in Table 3.

C. Parameter Estimates

Parameter estimates for each independent flight maneuver were determined using ordinary least-squares regression in an equation-error formulation in the time domain. Each parameter estimate for an individual maneuver was weighted based on its uncertainty, which was calculated from

$$w_i = \frac{1}{s_i^2} \quad (36)$$

where s_i is the standard error of each parameter estimate for an individual maneuver accounting for colored residuals [36]. The final parameter estimates were obtained from a weighted mean of the individual parameter estimates:

$$\hat{\theta} = \frac{\sum_{i=1}^n w_i \hat{\theta}_i}{\sum_{i=1}^n w_i} \quad (37)$$

and the final uncertainty for each parameter estimate is obtained from a propagation-of-errors formula:

$$\sigma = \left(\sum_{i=1}^n w_i \right)^{-1/2} \quad (38)$$

where n is the number of maneuvers, $\hat{\theta}_i$ is the parameter estimate from each individual maneuver, and w_i is the weighting calculated in Eq. (36) [37,38]. An estimate of the 95% confidence interval on the weighted mean is given by $\pm 2\sigma$. Parameter estimates with uncertainty values for the nominal glide and spin models are shown in Tables 4 and 5, respectively. Of note, although noisy regressor data are known to produce biased parameter estimates [3,39], this was mitigated by smoothing the regressor data in the time domain during

Table 4 Nominal glide model

	$\hat{\theta} \pm \sigma$
C_{x_a}	$+0.286 \pm 0.013$
C_{x_q}	$+1.426 \pm 0.232$
$C_{x_{\delta_e}}$	$+0.146 \pm 0.009$
$C_{x_{(\alpha^2)}}$	$+2.635 \pm 0.087$
$C_{x_{(\beta^2)}}$	$+0.233 \pm 0.013$
C_{x_o}	-0.075 ± 0.001
C_{y_β}	-0.576 ± 0.006
C_{y_p}	-0.505 ± 0.027
C_{y_r}	$+0.412 \pm 0.019$
$C_{y_{\delta_a}}$	-0.364 ± 0.014
$C_{y_{\delta_r}}$	$+0.115 \pm 0.003$
C_{y_o}	$+0.006 \pm 0.001$
C_{z_a}	-4.247 ± 0.057
C_{z_q}	-9.055 ± 1.200
$C_{z_{\delta_e}}$	-0.607 ± 0.052
$C_{z_{(\beta^2)}}$	-0.568 ± 0.084
C_{z_o}	-0.308 ± 0.007
C_{l_β}	-0.085 ± 0.002
C_{l_p}	-0.602 ± 0.009
C_{l_r}	$+0.132 \pm 0.006$
$C_{l_{\delta_a}}$	-0.371 ± 0.005
C_{l_o}	-0.002 ± 0.000
C_{m_a}	-0.587 ± 0.013
C_{m_q}	-9.006 ± 0.233
$C_{m_{\delta_e}}$	-0.742 ± 0.010
$C_{m_{(\alpha^2)}}$	-1.629 ± 0.157
$C_{m_{(\alpha^3)}}$	-2.481 ± 1.038
$C_{m_{(\alpha^3)}}$	-0.132 ± 0.000
C_{m_o}	$+0.011 \pm 0.001$
C_{n_β}	$+0.055 \pm 0.001$
C_{n_p}	$+0.047 \pm 0.004$
C_{n_r}	-0.122 ± 0.002
$C_{n_{\delta_a}}$	$+0.074 \pm 0.002$
$C_{n_{\delta_r}}$	-0.046 ± 0.001
$C_{n_\beta^2}$	$+0.005 \pm 0.006$
$C_{n_\beta^3}$	$+0.215 \pm 0.027$
C_{n_o}	$+0.002 \pm 0.000$

Table 5 Spin model

	$\hat{\theta} \pm \sigma$		$\hat{\theta} \pm \sigma$
C_{x_a}	$+0.055 \pm 0.011$	$C_{l_{\delta_a}}$	$+4.113 \pm 0.179$
$C_{x_{\dot{\alpha}}}$	-3.881 ± 0.213	C_{l_β}	-0.058 ± 0.006
C_{x_β}	-0.018 ± 0.009	$C_{l_{\dot{\beta}}}$	$+0.205 \pm 0.032$
C_{x_Ω}	$+0.406 \pm 0.009$	C_{l_Ω}	-0.075 ± 0.006
$C_{x_{p_{osc}}}$	$+0.116 \pm 0.044$	$C_{l_{p_{osc}}}$	-0.040 ± 0.027
$C_{x_{q_{osc}}}$	$+4.801 \pm 0.242$	$C_{l_{q_{osc}}}$	-3.474 ± 0.190
$C_{x_{r_{osc}}}$	-0.116 ± 0.029	$C_{l_{r_{osc}}}$	$+0.304 \pm 0.037$
$C_{x_{\delta_e}}$	$+0.093 \pm 0.007$	$C_{l_{\delta_e}}$	$+0.029 \pm 0.004$
$C_{x_{\delta_a}}$	-0.040 ± 0.008	$C_{l_{\delta_a}}$	-0.094 ± 0.006
$C_{x_{\delta_r}}$	-0.027 ± 0.004	$C_{l_{\delta_r}}$	-0.018 ± 0.004
$C_{x_{(\dot{\alpha}^2)}}$	$+0.229 \pm 0.054$	$C_{l_{(\dot{\alpha}^2)}}$	$+0.208 \pm 0.033$
C_{x_o}	-0.125 ± 0.001	C_{l_o}	$+0.007 \pm 0.001$
$C_{y_{\dot{\alpha}}}$	$+12.302 \pm 0.546$	C_{m_a}	-0.140 ± 0.035
C_{y_β}	-0.472 ± 0.020	$C_{m_{\dot{\alpha}}}$	$+16.148 \pm 0.794$
$C_{y_{\dot{\beta}}}$	$+0.857 \pm 0.098$	C_{m_β}	-0.055 ± 0.026
C_{y_Ω}	$+0.025 \pm 0.014$	C_{m_Ω}	$+0.075 \pm 0.029$
$C_{y_{p_{osc}}}$	-0.139 ± 0.104	$C_{m_{p_{osc}}}$	$+0.399 \pm 0.125$
$C_{y_{q_{osc}}}$	-6.622 ± 0.575	$C_{m_{q_{osc}}}$	-16.315 ± 0.862
$C_{y_{r_{osc}}}$	$+0.953 \pm 0.108$	$C_{m_{r_{osc}}}$	$+0.493 \pm 0.098$
$C_{y_{\delta_e}}$	-0.083 ± 0.015	$C_{m_{\delta_e}}$	-0.120 ± 0.024
$C_{y_{\delta_a}}$	-0.006 ± 0.013	$C_{m_{\delta_a}}$	$+0.050 \pm 0.026$
$C_{y_{\delta_r}}$	$+0.070 \pm 0.010$	$C_{m_{\delta_r}}$	-0.054 ± 0.014
$C_{y_{(\dot{\alpha}^2)}}$	-0.774 ± 0.112	$C_{m_{(\dot{\alpha}^2)}}$	-0.531 ± 0.163
C_{y_o}	-0.063 ± 0.002	$C_{m_{(\delta_e \delta_a)}}$	-0.376 ± 0.146
C_{z_a}	-0.068 ± 0.051	C_{m_o}	$+0.005 \pm 0.003$
$C_{z_{\dot{\alpha}}}$	$+68.789 \pm 1.291$	C_{n_a}	$+0.045 \pm 0.003$
C_{z_β}	-0.261 ± 0.041	$C_{n_{\dot{\alpha}}}$	-0.669 ± 0.057
C_{z_Ω}	$+0.243 \pm 0.042$	C_{n_β}	$+0.016 \pm 0.002$
$C_{z_{p_{osc}}}$	$+2.160 \pm 0.186$	$C_{n_{\dot{\beta}}}$	-0.047 ± 0.011
$C_{z_{q_{osc}}}$	-65.041 ± 1.353	C_{n_Ω}	$+0.033 \pm 0.002$
$C_{z_{r_{osc}}}$	$+0.580 \pm 0.136$	$C_{n_{p_{osc}}}$	$+0.049 \pm 0.011$
$C_{z_{\delta_e}}$	$+0.159 \pm 0.041$	$C_{n_{q_{osc}}}$	$+0.524 \pm 0.061$
$C_{z_{\delta_a}}$	$+0.042 \pm 0.038$	$C_{n_{r_{osc}}}$	-0.169 ± 0.013
$C_{z_{\delta_r}}$	-0.062 ± 0.020	$C_{n_{\delta_e}}$	-0.002 ± 0.001
$C_{z_{(\dot{\alpha}^2)}}$	-2.024 ± 0.266	$C_{n_{\delta_a}}$	$+0.038 \pm 0.004$
$C_{z_{(\delta_e \delta_a)}}$	-0.396 ± 0.232	$C_{n_{\delta_r}}$	-0.041 ± 0.001
C_{z_o}	-0.856 ± 0.004	$C_{n_{(\dot{\alpha}^2)}}$	-0.048 ± 0.019
$C_{l_{\dot{\alpha}}}$	-0.026 ± 0.008	C_{n_o}	$+0.000 \pm 0.000$

the data processing step and before determining the model structure or estimating parameters.

D. Validation

Model validation aims to demonstrate that an adequate model was developed by testing the predictive capability using data not used for model identification [3,26,29]. Validation seeks to determine if 1) the model agrees sufficiently well with observed data and 2) the model is adequate for the intended purpose [26]. For this effort, validation focused on the first issue by comparing the model predictions to flight data from maneuvers excited with an independent validation multi-sine. The second issue is of primary importance for the model use, as outlined in Ref. [2] for this case, and is adequately addressed when the problem that motivated the modeling effort is successfully resolved. To answer the question of whether the model is good enough, two primary metrics were used. For quantitative analysis, similar to Refs. [9,10,40], the Theil inequality coefficient (TIC) and normalized root-mean-square error (NRMSE) were determined for

each model relative to a validation maneuver. The TIC is calculated from

$$\text{TIC} = \frac{\sqrt{\frac{1}{N}(\mathbf{z} - \mathbf{y})^T(\mathbf{z} - \mathbf{y})}}{\sqrt{\frac{1}{N}\mathbf{z}^T\mathbf{z} + \frac{1}{N}\mathbf{y}^T\mathbf{y}}} \quad (39)$$

where $\mathbf{z}$ is the force or moment coefficient (C_x , C_y , C_z , C_l , C_m , or C_n) calculated from measured flight data, and $\mathbf{y}$ is the model prediction of the respective force or moment coefficient. TIC values range from zero to one, with a TIC value of zero indicating a perfect model fit. According to Ref. [29], a TIC value below 0.3 is considered good agreement between the validation data and model predictions. However, TIC values are dependent on the root mean square (RMS) of the data, as seen in the denominator of Eq. (39), where low RMS values can bias TIC results high, and vice versa. Thus, the analyst

must use caution and consider additional metrics. Accordingly, the NRMSE is calculated for each force and moment coefficient from

$$\text{NRMSE} = \frac{1}{\text{range}(z)} \sqrt{\frac{(z - y)^T (z - y)}{N}} \quad (40)$$

where N is the number of samples in the dataset. A NRMSE value of zero indicates a perfect model fit, and the magnitude can be interpreted similar to a percent error. Range normalization provides a fair assessment of the error for coefficients such as C_x and C_z , which are not centered about zero [40].

Table 6 Nominal glide maneuver validation metrics^a

Model basis	C_x	C_y	C_z	C_l	C_m	C_n
<i>Theil inequality coefficient</i>						
Nominal glide	0.088	0.087	0.102	0.228	0.235	0.103
<i>Normalized root-mean-square error</i>						
Nominal glide	0.085	0.047	0.093	0.084	0.095	0.045

^a0 indicates perfect model fit.

Table 7 Spin maneuver model validation metrics^a

Model basis	C_x	C_y	C_z	C_l	C_m	C_n
<i>Theil inequality coefficient</i>						
Nominal glide	0.326	0.326	0.152	0.710	0.779	0.564
Spin	0.027	0.118	0.017	0.372	0.368	0.291
<i>Normalized root-mean-square error</i>						
Nominal glide	1.032	0.373	0.640	0.461	0.799	0.395
Spin	0.078	0.123	0.068	0.126	0.115	0.138

^a0 indicates perfect model fit.

The nonlinear model predictions for each flight condition (nominal glide and spin) were validated against flight data from maneuvers excited with a set of independent validation multisine inputs that were designed with different power spectra than the multisine design used for modeling. This prediction test, as described in Refs. [3,29], duly challenges the model by using data that were not included in the modeling process and were obtained using a different excitation. Three validation cases are considered here for comparison:

- 1) The nominal glide model is validated against an independent validation multisine maneuver in a nominal glide.
- 2) The nominal glide model is validated against an independent validation multisine spin maneuver.
- 3) The spin model is validated against an independent validation multisine spin maneuver.

For the nominal glide validation maneuver, quantitative validation results are shown in Table 6. For the spin validation maneuver, quantitative validation results are shown in Table 7. Figures 8 and 9 show the validation results for a nominal glide maneuver and a spin maneuver with the independent validation multisine, along with the smoothed flight data. The same flight data are used for both the nominal glide model and spin model validation in Fig. 9 to directly compare model performance in a spin.

For the nominal glide model validated against a nominal glide maneuver, the TIC values are much lower than 0.3, indicating a good fit for the data. Additionally, the low NRMSE values demonstrate adequate model performance. For the spin model validated against a spin maneuver, the low TIC and NRMSE values also indicate adequate model performance. Visual inspection of nominal glide model predictions to nominal glide validation data (Fig. 8) and spin model predictions to spin validation data (Fig. 9b) reveals good matching between the flight data and the model predicted force and moment coefficients.

On the contrary, the nominal glide model does not validate well against the spin maneuver. Relatively high TIC and NRMSE values indicate poor model performance. Visual inspection of the model-predicted force and moment coefficients in Fig. 9a reveals that the nominal glide model predicts exaggerated aircraft response while the

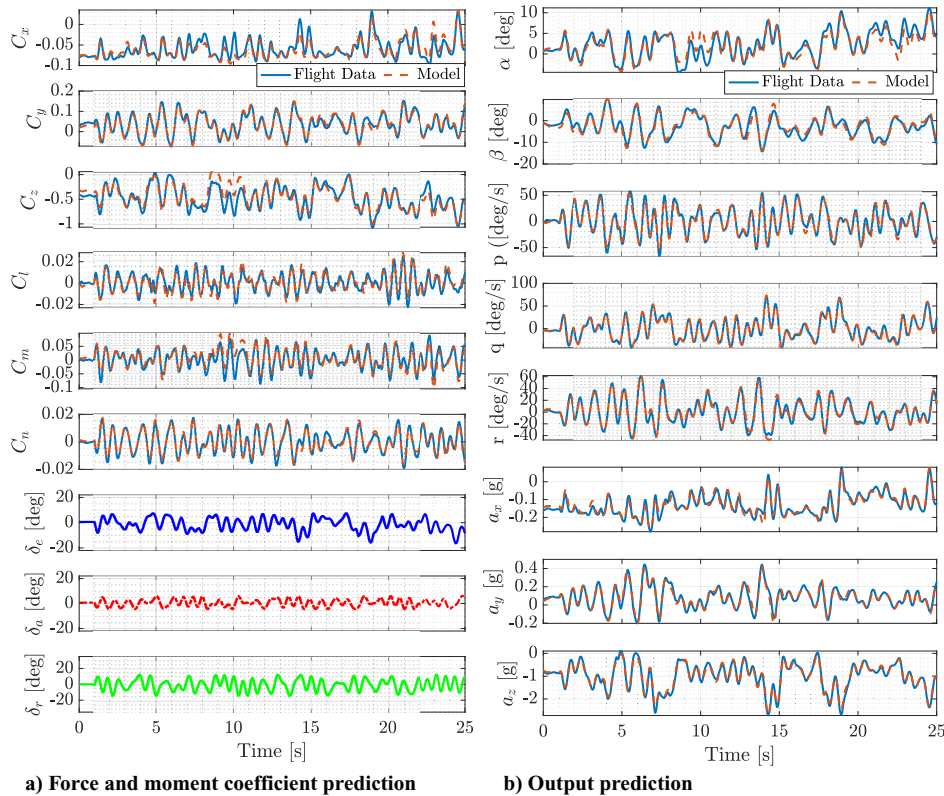


Fig. 8 Validation results for nominal glide model for nominal glide maneuver with multisine inputs.

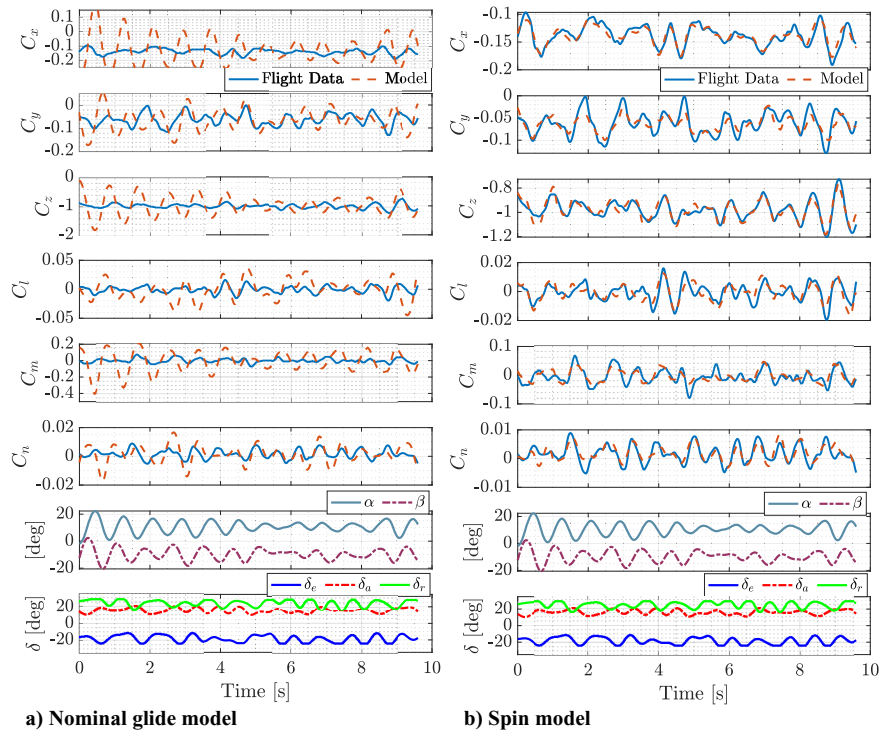


Fig. 9 Comparison of validation results for a spin maneuver with multisine inputs (note the same spin flight data with different scales).

aircraft is in a spin. This indicates the nominal glide model is not a good predictor of spin dynamics.

V. Conclusions

In the investigation described here, a nonlinear, quasi-steady, coupled aerodynamic model was developed for a fixed-wing aircraft in a nominal glide as well as a spinning descent. New methods were proposed to adequately describe the aircraft dynamics and control authority in the neighborhood of a stable, oscillatory spin. Orthogonal phase-optimized multisine inputs were used as excitation signals while collecting spinning flight data, and a new decomposition of explanatory variables led to an effective model structure for spin flight data analysis. The results show that the nominal glide flight dynamic model predictions matched validation data well for conventional flight, although not for a spinning descent. Predictions using the model developed for spinning flight, on the other hand, adequately matched validation data for a spinning descent excited with independent multisine inputs.

A model-based spin path control law developed using the more accurate spinning flight model should perform better than one developed using the nominal glide flight model. As illustrated here, however, developing a spinning flight model is not a trivial exercise due to the nonlinear, unsteady, and coupled aerodynamics, including the necessity to place the aircraft in a stall spin for model development. A parallel research effort in Ref. [2] demonstrated an effective robust control law designed using the nominal envelope gliding flight dynamic model developed in this paper to adjust the flight path while in a spinning descent. Future efforts aim to improve on flight path control for an aircraft in a terminal spin using the methods and spinning flight model developed in this paper. Regardless of the model used for spin control design, the spinning flight model can be used to assess and tune performance in simulations in advance of flight tests involving closed-loop control.

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